

Discussed guidelines with contractor team & site team



- ❑ **Pipe Cutting**: Cut the pipe straight to the required length.
- ❑ **Chamfering**: Chamfer every cut pipe using a file.
- ❑ **Hole Cutting**: Use a 44mm hole saw cutter to make holes for the Multi-Floor trap.
- ❑ **Lubrication**: Apply Huliot lubricant to both the pipe and fitting before fixing them together. 🛢️
- ❑ **Slope** : Maintain a slope of 1:100 MM 📐
- ❑ **Avoid Heating** Do not heat the pipe to create the socket, and do not make a hole to trap by heating 🚫🔥

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- ❑ **Lubrication**: Apply Huliot lubricant to both the pipe and fitting before fixing them together. 🛢️
- ❑ **Maintenance Gap**: Maintain a distance of 15mm to 20mm between the slab and fitting for future maintenance.
- ❑ **Bends**: Use two 45-degree bends below the WC and washbasin instead of a 90-degree bend.
- ❑ **Clamping**: Fix a clamp near the pipe socket and at every change in direction.
- ❑ **Trap Support**: Every trap needs channel support. 🔧
- ❑ **Fasteners**: Use 10mm rods with bullet fastener nuts and bolts. 🛡️
- ❑ **Slope** : Maintain a slope of 1:100 MM 📐

❑ **Avoid Heating** Do not heat the pipe to create the socket, and do not make a hole to trap by heating 🚫🔥

Create a mock-up for both vertical and internal

General point

- Eccentric reducer Installation :- in a horizontal drainage line, place the eccentric reducer with the flat part at the bottom
- Open pipe ends and trap top needs to close to avoid any cement slurry , cement mortar not goes inside trap. Trap needs to clean and water seal needs to fix properly before fixing zali to avoid any foul smell issue later stage.
- In the 75 × 75 mm Y-fitting, the 75 mm branch connected through the 45° angle shall be kept slightly higher than the straight branch. This arrangement helps prevent reverse flow from entering the trap
- Connect the pipe and fitting freely without applying any stress
- In vertical lines, use a door bend instead of a plain bend for maintenance purposes.
- In a single stack drainage system, only SWIFT fittings are recommended as per standard practice. The use of Y-fittings is not advised, as they can adversely affect airflow and pressure balance within the stack.
- Wherever there is a change in pipe direction, two 45° bends shall be provided in place of a sharp 90° bend to achieve smoother flow and reduce flow resistance.

General point

- In a single stack drainage system, only SWIFT fittings are recommended as per standard practice. The use of Y-fittings is not advised, as they can adversely affect airflow and pressure balance within the stack.
- At site, PVC multi floor traps are being used in certain locations. PVC and PP are different materials and follow different jointing methodologies, hence mixing both systems is not recommended. PVC floor traps do not provide a 50 mm water seal, which is a non-compliance with Single Stack System standards and may result in water seal loss under negative pressure conditions. Huliott PP multi traps are installed from the 2nd to 30th floor and are compliant with the system requirements, whereas PVC floor traps are installed from the 31st to 33rd floor, resulting in inconsistency in material usage and system performance across the stack.
- Huliott always recommended to execute the work strictly as per approved drawings and specifications. Any change in material, system, or installation methodology should not be carried out without prior consultation and approval from the consultant and other key project stakeholders.

General Point

- **Open pipe ends and trap top needs to close to avoid any cement slurry , cement mortar not goes inside trap. Trap needs to clean and water seal needs to fix properly before fixing zali to avoid any foul smell issue later stage.**
- **Execution of the vertical mock-up is essential. As the shaft contains multiple services, clearance within the shaft must be checked. Any clearance or alignment issues identified after completion of internal works are not rectifiable.**
- **All pipes and fittings must be fixed at the correct angle.**
- **Huliot always recommends that all installations be carried out strictly as per approved drawings. If any site condition requires a change, prior approval from the concerned consultant must be obtained before execution.**

vertical mock-up, as the shaft contains multiple services and adequate clearance within the shaft must be checked before proceeding with further works

General Point PRL and Diversion

- The PRL and main header line should not be connected at the same level. The PRL connection must be positioned above the main header line to ensure proper flow.
- Provide an inspection pipe at the end of the PRL, and then connect it to the main header line.
- Pipe support spacing exceeds 600 mm. MS/GI rigid support to be provided accordingly Huliott recommends providing M.S./G.I. rigid supports to secure PP push-fit pipes. The suggested spacing for rigid supports is 1.5 times the pipe diameter. (e.g., For 110 mm pipe: $110 \text{ mm} \times 1.5 = 1.65 \text{ meters}$). Rigid supports must also be provided at every change in direction.
- End lock /lock seal should be provided at the end point of the line.

General Point Diversion

- **Pipe support spacing exceeds 600 mm. MS/GI rigid support to be provided accordingly**
- **Use Chilli Clamp (Round Rabarise) to provide proper grip, reduce vibration and noise, prevent pipe surface damage, and ensure long-term pipe stability**
- **Lock seals shall be fitted in front of the end caps to ensure proper sealing of the system**
- **Support/clamp should not be fixed directly to the fitting. The support shall be installed on the pipe section adjacent to the fitting.**

General Point Vertical

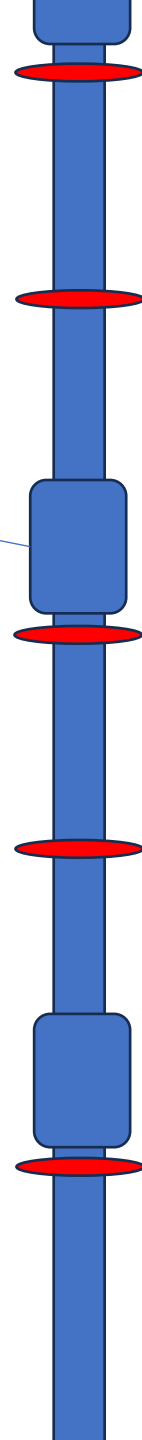
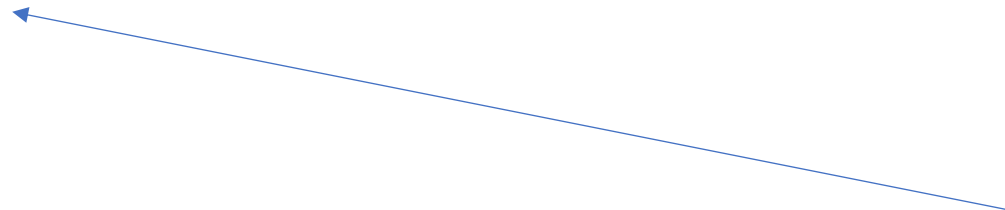
Do not insert the fitting into the wall. Maintain a minimum 10-15 mm gap between the fitting and the wall to allow proper installation, movement, and maintenance clearance.

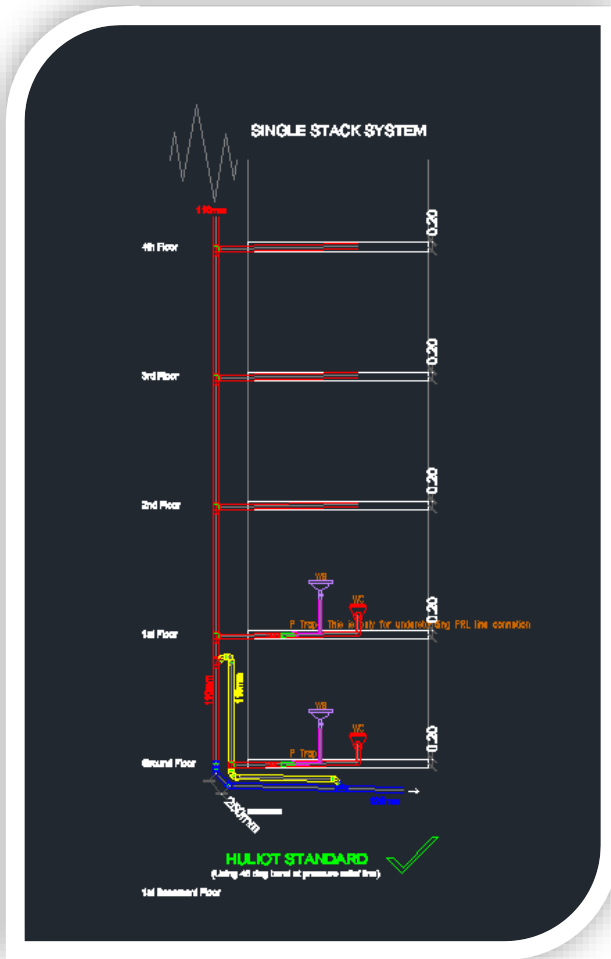
Near pipe socket clamp need to fix.

Clamp need to fix every changing direction

Huliot always recommends that all installations be carried out strictly as per approved drawings. If any site condition requires a change, prior approval from the concerned consultant must be obtained before execution.







Observation / Requirement

PRL clearance is a critical installation requirement. As per standard installation guidelines, a minimum vertical and horizontal distance of 2.4 meters shall be maintained for PRL.

Non-Compliance / Caution

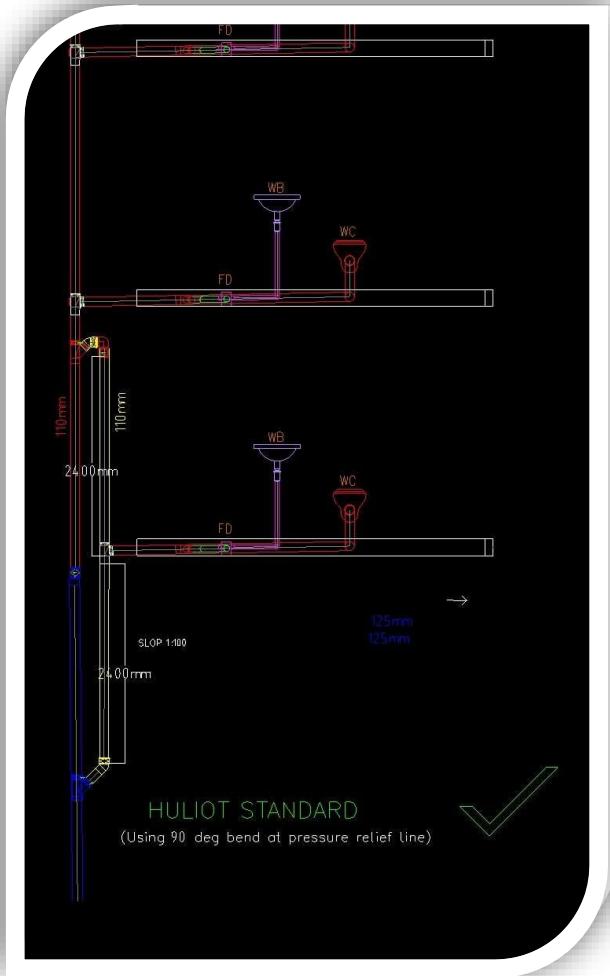
Any PRL distance less than 2.0 meters is not acceptable and may affect system performance and safety.

Instruction

Ensure the PRL is installed by maintaining the recommended 2.4 m clearance. Under no circumstances should the distance be reduced below the minimum 2.0 m requirement.

Why PRL Is Important

PRL is important to ensure proper airflow and pressure balance in the drainage system. Maintaining the required PRL distance prevents negative pressure, protects trap seals, and allows smooth discharge of wastewater. Insufficient PRL distance can lead to trap seal loss, foul odor, and improper system performance.



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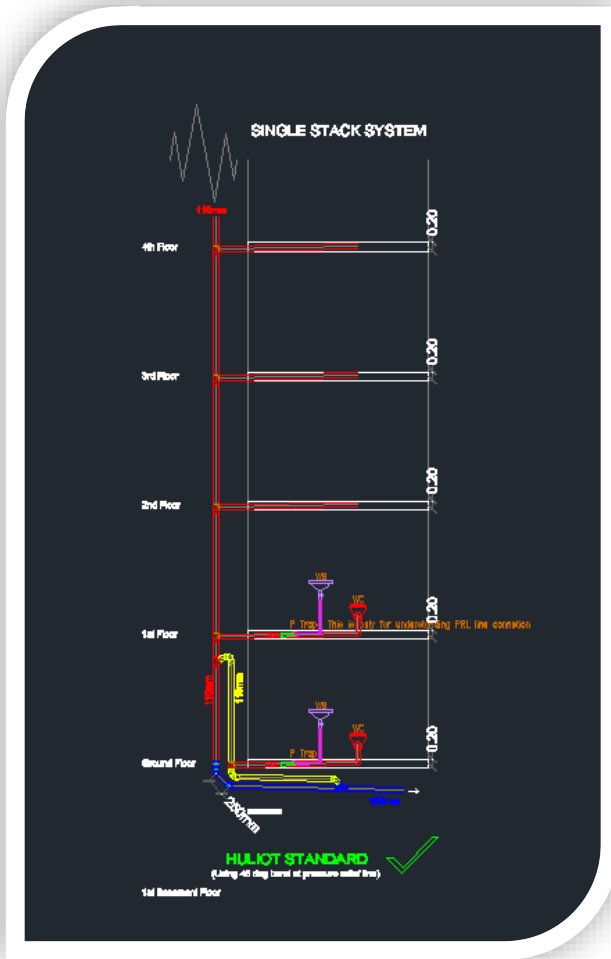
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Standard Reference:

- As per **EN 12056 (Part 2)** – Design of gravity drainage systems inside buildings
- Pressure fluctuation control is mandatory in single stack systems to maintain trap seal integrity
- In Huliote system design also, PRL is recommended.

Why PRL Is Important

PRL is important to ensure proper airflow and pressure balance in the drainage system. Maintaining the required PRL distance prevents negative pressure, protects trap seals, and allows smooth discharge of wastewater. Insufficient PRL distance can lead to trap seal loss, foul odor, and improper system performance.

In the Shaft outlet pipe is not straight, vertical fittings will not seat properly, leading to poor joints. Fix outlet pipe straight. Before waterproofing.



Trap should be installed in straight and level position. Do not tilt the trap, as tilting can reduce/affect the required water seal and may lead to foul smell or improper trap performance. Correct the trap alignment immediately.

Trap installation is not acceptable. The trap must be fixed straight and level to maintain the required minimum water seal. Tilting of the trap may disturb the water seal and affect trap performance. Rectification is required immediately.

Huliot single stack requirement mentions **minimum water seal ≥ 50 mm** for proper system performance.



Why Huliote US pipe is Black

Absorption vs. Penetration

While a white or light-colored pipe reflects some UV, those rays can still penetrate the surface and break the chemical bonds of the plastic deep inside, making the pipe brittle over time.

- The "Sunscreen" Effect:** By adding **Carbon Black** to the outer layer, Huliote ensures the UV rays are absorbed right at the very surface (the first few microns).
- Sacrificial Layer:** The carbon black particles act like a shield, turning the UV energy into a tiny amount of heat before it can reach the structural middle layer. It "sucks in" the rays so they can't go any deeper.

Real-Life Example: The Car Tire

Think about a **car tire**. Why is it always black? Tires stay outside in the hot sun all day, every year. If they were white or clear, the sun would destroy the rubber in a few weeks. Because they are black (with that same Carbon Black ingredient), they stay strong for years.

It is recommended to use a High Riser above the Multi Floor Trap instead of a plain pipe piece. The High Riser provides a proper push-fit connection with dual EPDM sealing rings, ensuring a watertight and secure joint between the trap and extension. Using a plain pipe piece may not provide reliable sealing, which can lead to leakage, movement, or maintenance issues during service life.

Why use a High Riser?

- ❑ Ensures proper sealing with two EPDM rings.
- ❑ Provides a secure connection between the trap and extension.
- ❑ Reduces the risk of water leakage.
- ❑ Maintains the designed performance of the Multi Floor Trap.
- ❑ Recommended as per the Huliote Multi Floor Trap installation arrangement, which includes a height riser above the trap.



USE HIGH RISER ABOVE TRAP – NOT PIPE PIECE

Use High Riser above the Multi Floor Trap instead of pipe piece for proper sealing and long-term leak proof performance.

✓ USE HIGH RISER

High Riser

Push-fit connection with 2 EPDM rings

Upper EPDM Ring

Seals between High Riser and Trap

Lower EPDM Ring

Seals inside the Trap socket

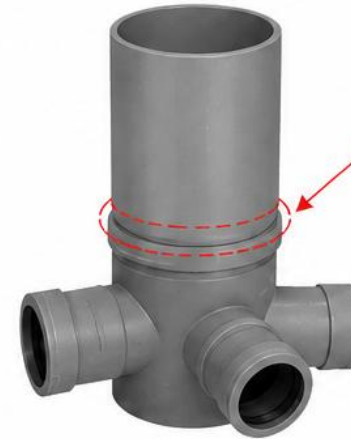


BENEFITS

- ✓ Proper sealing with 2 EPDM rings
- ✓ Leak proof connection
- ✓ Secure & reliable installation
- ✓ Recommended in Huliott installation arrangement



✗ DO NOT USE PIPE PIECE



No proper sealing between trap and pipe piece

ISSUES

- ✗ No EPDM ring sealing
- ✗ Risk of leakage
- ✗ Less secure connection
- ✗ Not recommended



Work done as per consultant drawing. **Close the open ends of trap and pipes to avoid entry of any particles at the time of other site activity.** Huliott suggest using a 50mm horizontal vent pipe as the length from WC fixture to the shaft is more than 04 meter.

Code Reference

EN 12056-2, Clause 6.3 – System III (Single Stack):

- WC branch (DN 100): max unventilated branch length ≤ 1.0 m without additional vent
- With ventilated branch: up to **4.0 m** is permissible (with vent pipe at WC end)
- Beyond 4.0 m: mandatory separate vent pipe – no exceptions**

NBC 2016, Part 9, Section 3 similarly restricts long unventilated soil branches.



Vertical mockup done as per drawing. don't install fitting inside the wall **Maintain gap of 15-20mm between the fitting and the wall.**



Vertical work has been executed as per site conditions, not as per the drawing.

The 110mm pipe outlet was not aligned in the same vertical line; therefore, the downtake line has been adjusted to maintain a straight alignment. This adjustment is technically acceptable.

110mm clamp need to fix red mark area.

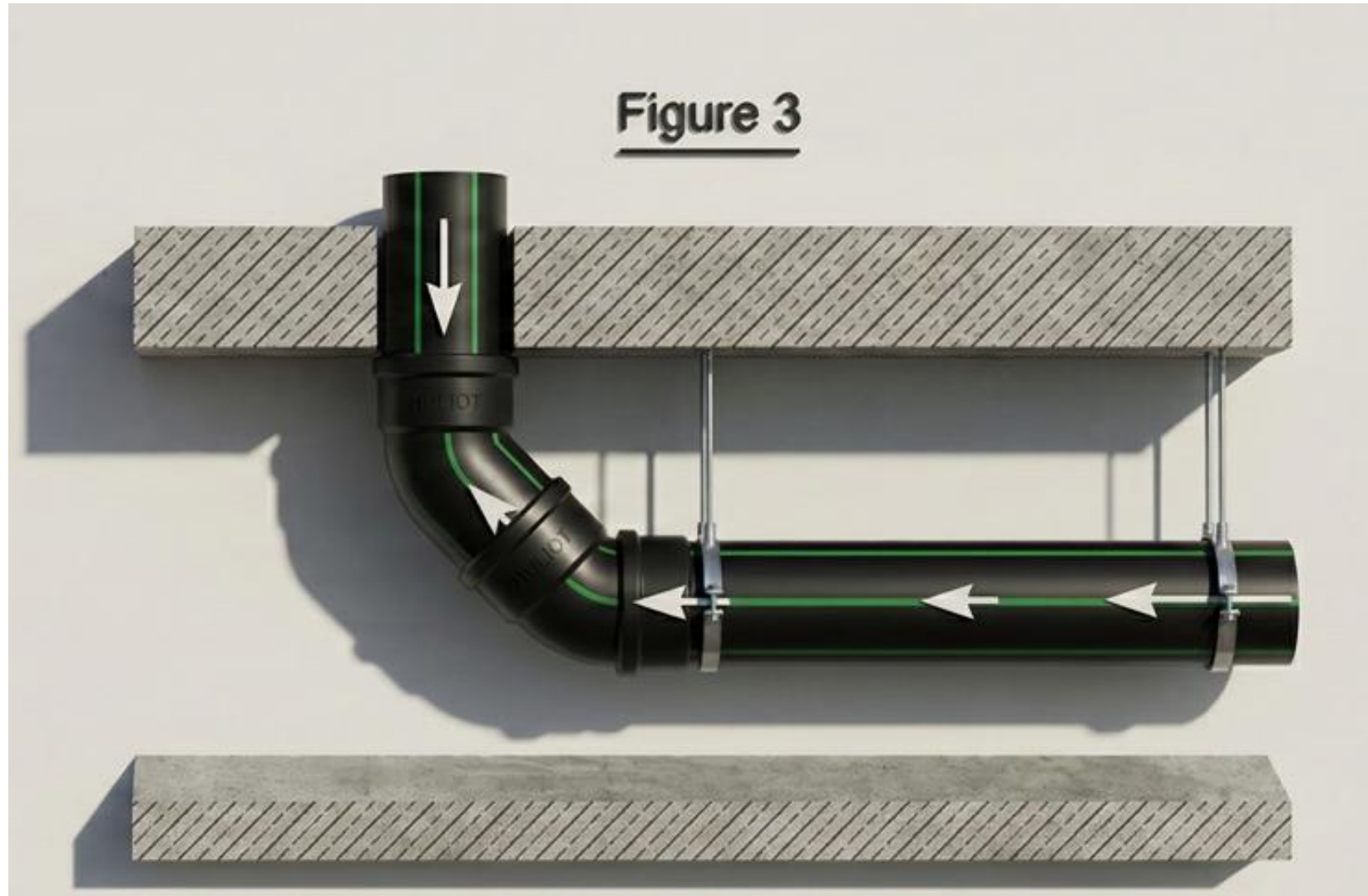


The system is functional; however, it is not as per the approved drawing.

- Technically it is acceptable, but the installation does not follow the consultant drawing.**
- The work is technically functional; however, it is not as per the approved drawing.**
- Technically acceptable, but not in compliance with the drawing.**

Installation has been done as per site condition. The contractor changed the washbasin pipe routine with approval from the site team. We suggest do as per drawing.

Clamp needs to be fixed from vertical to horizontal immediate turn, after 2x45 shoe, and at every change of direction.



It has been observed that fittings from other brands have been used in combination with Huliot fittings. This practice is not acceptable. Kindly ensure all mixed fittings are removed and replaced with Huliot fittings only.



Pipe support spacing exceeds 600 mm. MS/GI rigid support to be provided accordingly

Huliot recommends providing M.S./G.I. rigid supports to secure PP push-fit pipes. The suggested spacing for rigid supports is 1.5 times the pipe diameter. (e.g., For 110 mm pipe: $110 \text{ mm} \times 1.5 = 1.65 \text{ meters}$). Rigid supports must also be provided at every change in direction.



Pipe support spacing is observed to be more than 600 mm. Proper M.S./G.I. rigid supports shall be provided to securely fix the PP push-fit pipes.

As per recommendation, rigid support spacing should be maintained at 1.5 times the pipe diameter (for example, for 110 mm pipe – 1.65 m spacing). Additionally, rigid supports must be installed at every change in direction to ensure proper stability and alignment.

The PRL and main header line should not be connected at the same level. The PRL connection must be positioned above the main header line to ensure proper flow.

Provide an inspection pipe at the end of the PRL, and then connect it to the main header line.



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In a Single stack system minimum 50mm water seal is mandatory. The PVC trap currently installed on-site does not meet this requirement

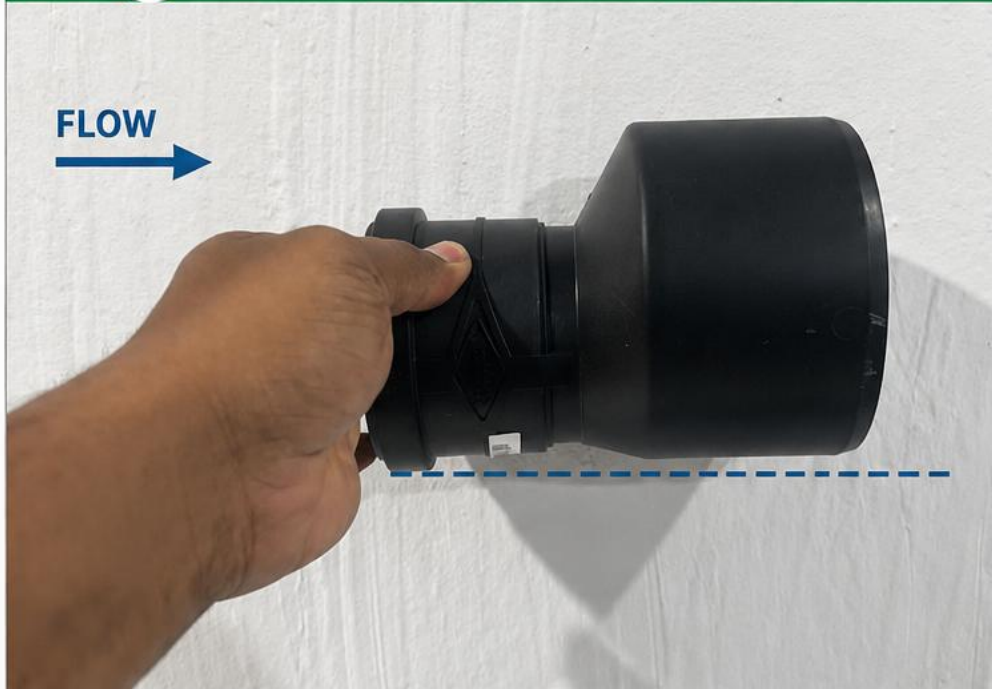
Eccentric reducer Installation :- in a horizontal drainage line, place the eccentric reducer with the flat part at the bottom



ECENTRIC REDUCER – RIGHT INSTALLATION POSITIONING



CORRECT: FLAT ON BOTTOM (FOB)



- ✓ Flat side on the bottom.
- ✓ Maintains a continuous invert.
- ✓ Prevents pooling and clogs.
- ✓ Correct for drainage / gravity lines.



INCORRECT: FLAT ON TOP



- ✗ Flat side on the top.
- ✗ Creates a step or pocket.
- ✗ Causes pooling, clogs, and odors.
- ✗ Incorrect for drainage / gravity lines.



REMEMBER: For horizontal drainage lines, always install the eccentric reducer **FLAT ON BOTTOM (FOB)** to maintain a continuous invert and prevent problems.



Back entry WC requires a vertical drop. As observed on site, the WC is connected directly using a sweep tee. Refer to the attached image and rectify the same immediately.

The back entry WC requires a proper vertical drop connection. However, as observed on site, the WC has been connected directly using a sweep tee.

This is not recommended. A vertical drop must be provided as per standard practice.

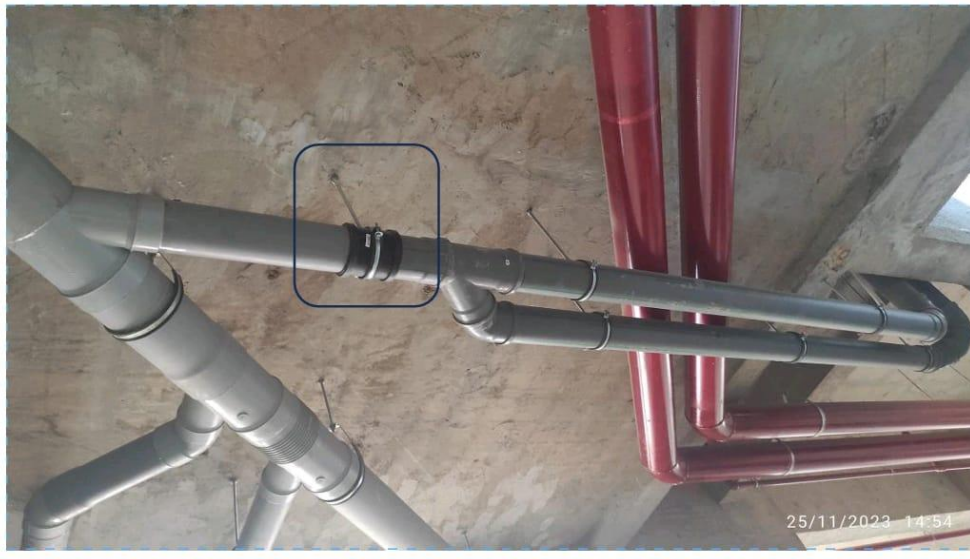
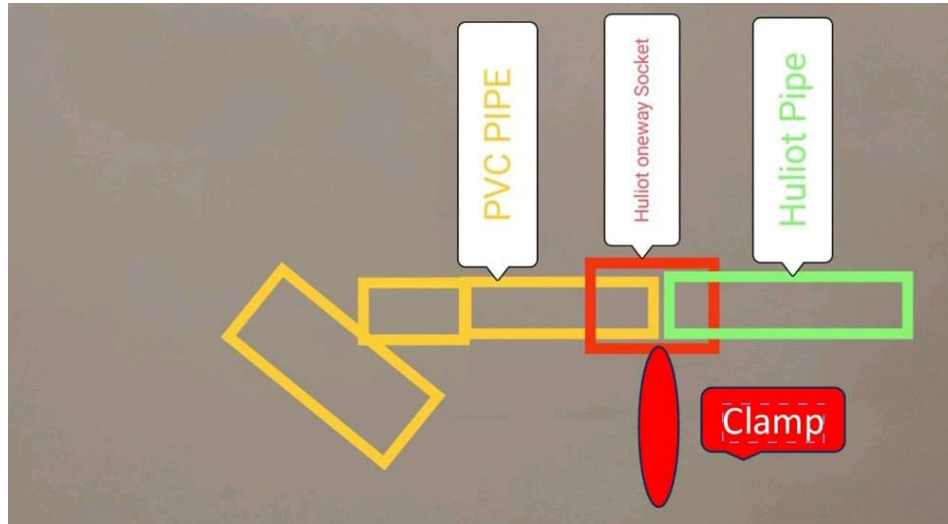
If the drop is not executed, there is a high risk of backflow issues at a later stage.

Kindly refer to the attached image and rectify the same immediately to avoid future operational problems.

SH 00 Back entry WC requires a proper vertical drop connection. However, as observed on site, the WC has been connected directly using a sweep tee, which is not as per standard practice. A vertical drop must be provided to ensure smooth discharge and prevent future issues. If the drop is not executed, there is a high risk of backflow problems at a later stage. Kindly refer to the attached image and rectify the same immediately to avoid operational issues.

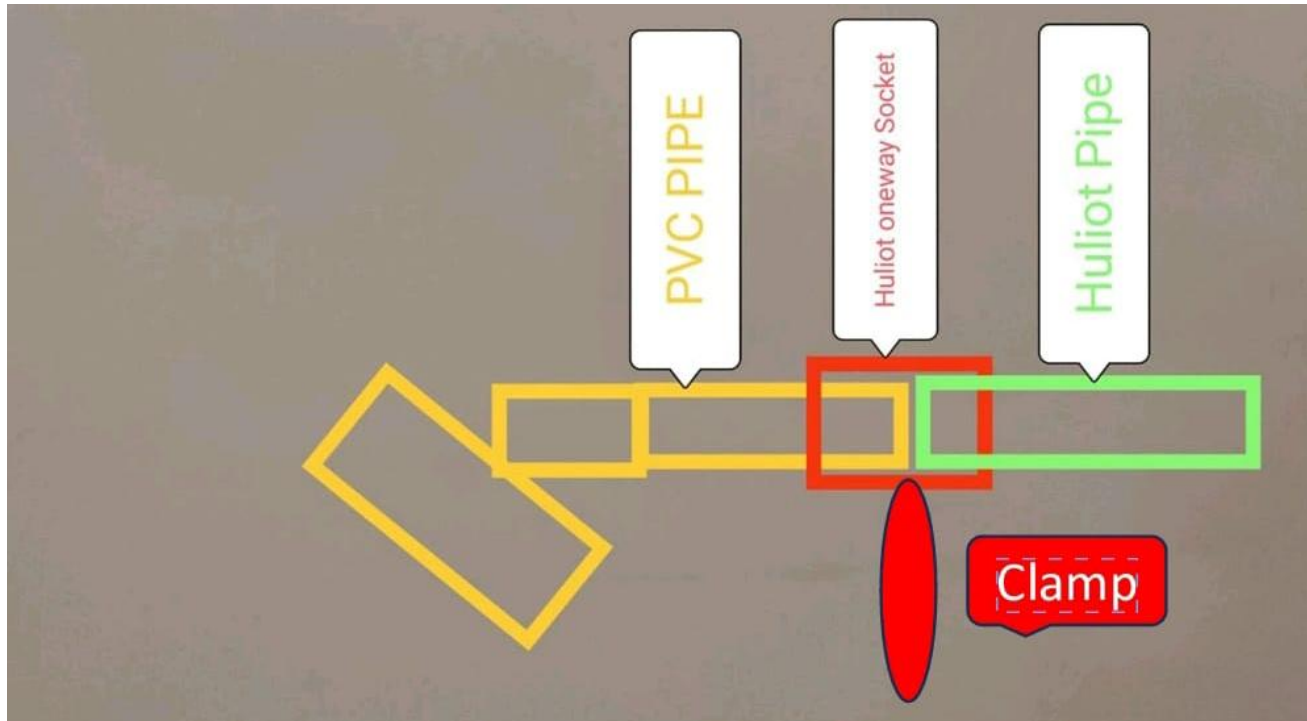


Huliot to PVC joint with one way socket



Huliot to PVC joint with one way socket

Do not joint PP and PVC directly. They are different materials, solvent cement does not work on PP, and both expand differently with temperature. This causes leakage and joint failure. Always use either full PP line or full PVC line.



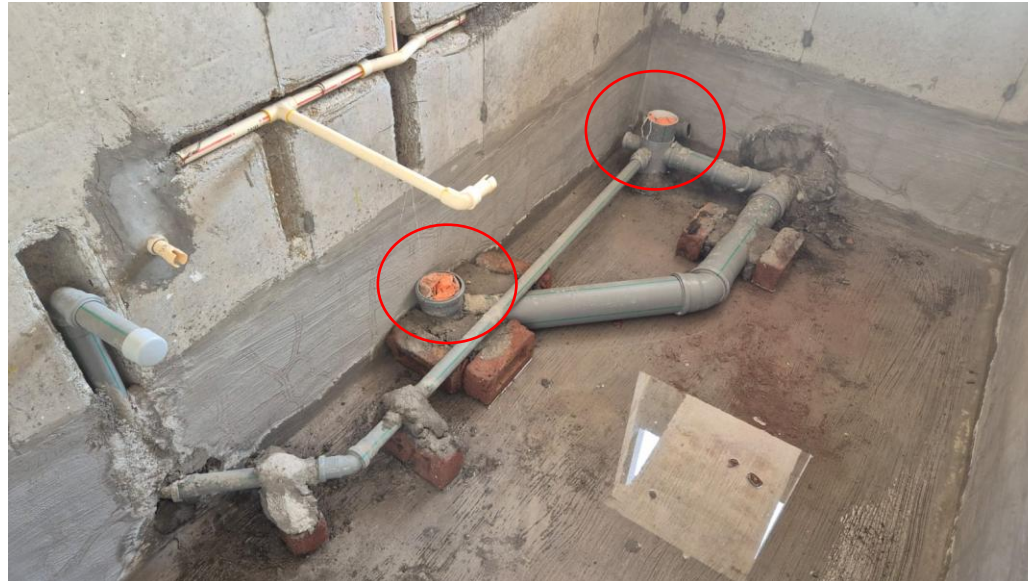
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Open pipe ends and trap top needs to close to avoid any cement slurry , cement mortar not goes inside trap. Trap needs to clean and water seal needs to fix properly before fixing zali to avoid any foul smell issue later stage.

All open pipe ends and trap tops must be securely sealed before waterproofing activities begin. This prevents cement slurry, mortar, debris, or chemicals from entering the drainage system and causing blockage, trap seal damage, or long term performance issues

All open pipe ends and trap tops should be properly covered during waterproofing work so that cement slurry, mortar, or debris does not enter inside. This prevents blockage, damage to the trap seal, and future drainage problems.



Oversized clamps have been used at the site. Please replace them with the correct size clamps to ensure proper installation

During the visit, waterproofing work was already completed. The pipe was not visible, therefore the workmanship could not be inspected or verified.



75mm pipe alignment is not proper shift 110x75
back side and connect properly.



Kindly ensure a clear gap of 15–20 mm is maintained between the slab and the fitting. Request not to insert the fitting inside the core, as this will help facilitate smooth future maintenance.



Maintain a 15 to 20 mm gap between the slab and the fitting. Do not insert the fitting inside the core.



Don't install the clamp on the fitting. Fix the clamp only near the pipe socket.



50 mm pipe in stress observed. Connect the pipe and fitting freely without applying any stress

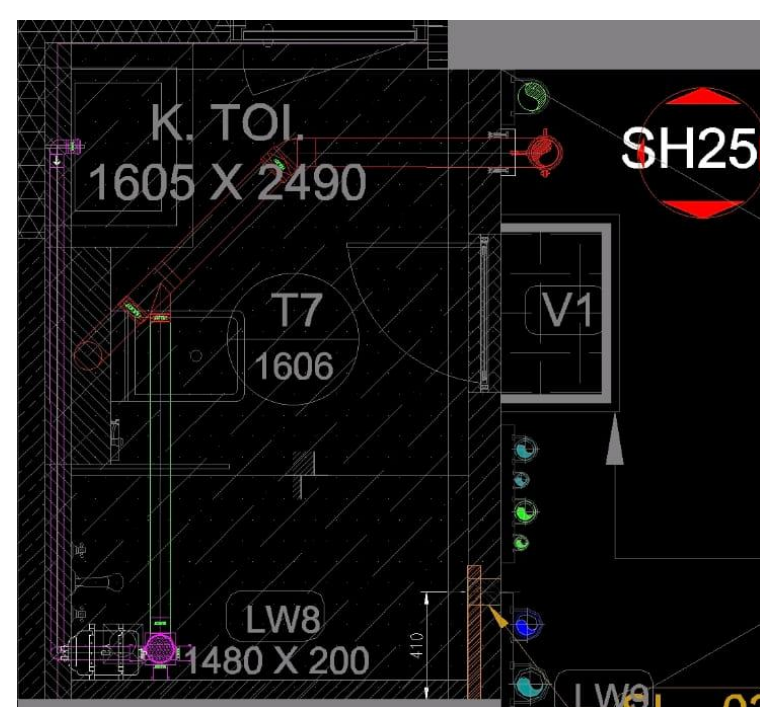


Do not connect the 110 × 75 Y fitting directly to the WC. Provide two 45° bends and maintain at least 300 mm pipe distance in between.

The 110 × 75 mm Y-connection is observed to be too close to the WC outlet pipe. A minimum distance of 300 mm shall be maintained between the WC outlet and the Y-connection.



Wash basin connection pending. Use height riser above multi floor trap. Multi trap connection to be connection 300mm away from the WC to fixture avoid future damage. Close open ends with end cap to avoid entry of any particles at the time of other site activity.



In the 110x75 Y fitting, keep the 75 mm branch slightly higher than the 110 mm branch to prevent WC discharge from entering the trap



The end cap has been installed incorrectly. First, install the end cap, and then fix the lock seal. Please correct this immediately.

Installation Methode :- First, install the end cap, and then fix the lock seal.



In a proper horizontal drainage installation, the access plug should always be positioned on the upper side (the top) of the pipe.





End cap and Lock seal

End Lock

End caps have been installed at the pipe end point. Lock seals need to be fixed in front of the end caps to ensure proper sealing of the system



Universal Clamp



Chilli Clamp



"Universal (U) Clamp has been used for pipe support. It is recommended to replace it with a rubber-lined Chilli Clamp to provide proper grip, reduce vibration and noise, prevent pipe surface damage, and ensure long-term pipe stability

Universal Clamp



Chilli Clamp



- Use Chilli Clamp (Round Rabarise) to provide proper grip, reduce vibration and noise, prevent pipe surface damage, and ensure long-term pipe stability

Key Benefits of Chilli Clamp

1. Better Pipe Grip

1. Rubber lining grips the pipe firmly.
2. Reduces slipping and unwanted movement.

2. Vibration Control

1. Absorbs vibration from water flow.
2. Prevents vibration transfer to walls, slabs, and supports.

3. Noise Reduction

1. Minimizes rattling and pipe noise.
2. Recommended where acoustic performance is important.

4. Pipe Protection

1. Prevents scratches and wear caused by metal-to-metal contact.

5. Long-Term Support Reliability

1. Maintains proper pipe alignment.
2. Reduces maintenance issues caused by pipe movement and abrasion.

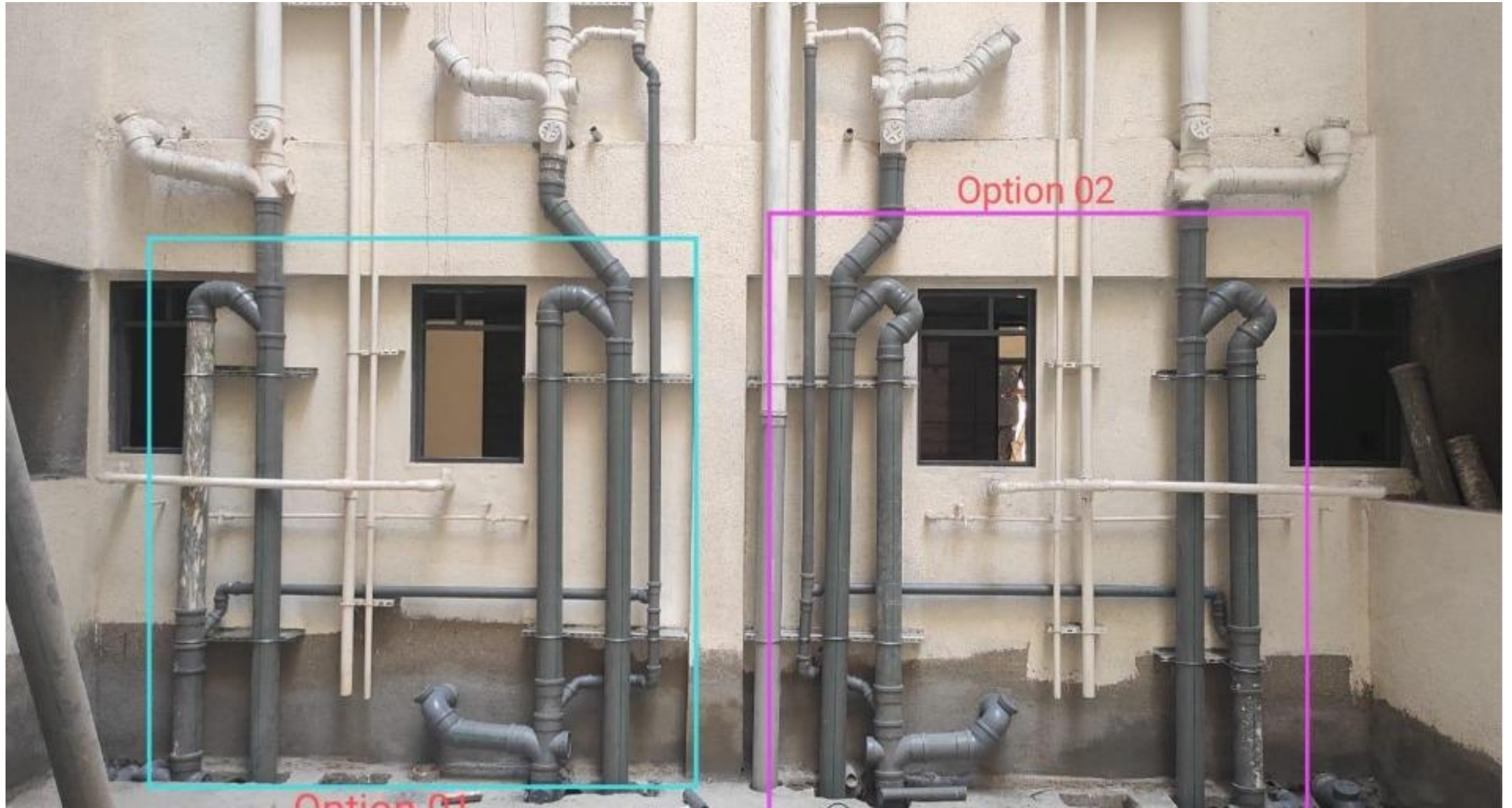
Chilli Clamp



Support/clamp should not be fixed directly to the fitting. The support shall be installed on the pipe section adjacent to the fitting.



PRL option



Observation:

Maintain a **250-300 mm straight pipe length** before diverting the drainage line with a bend or offset.

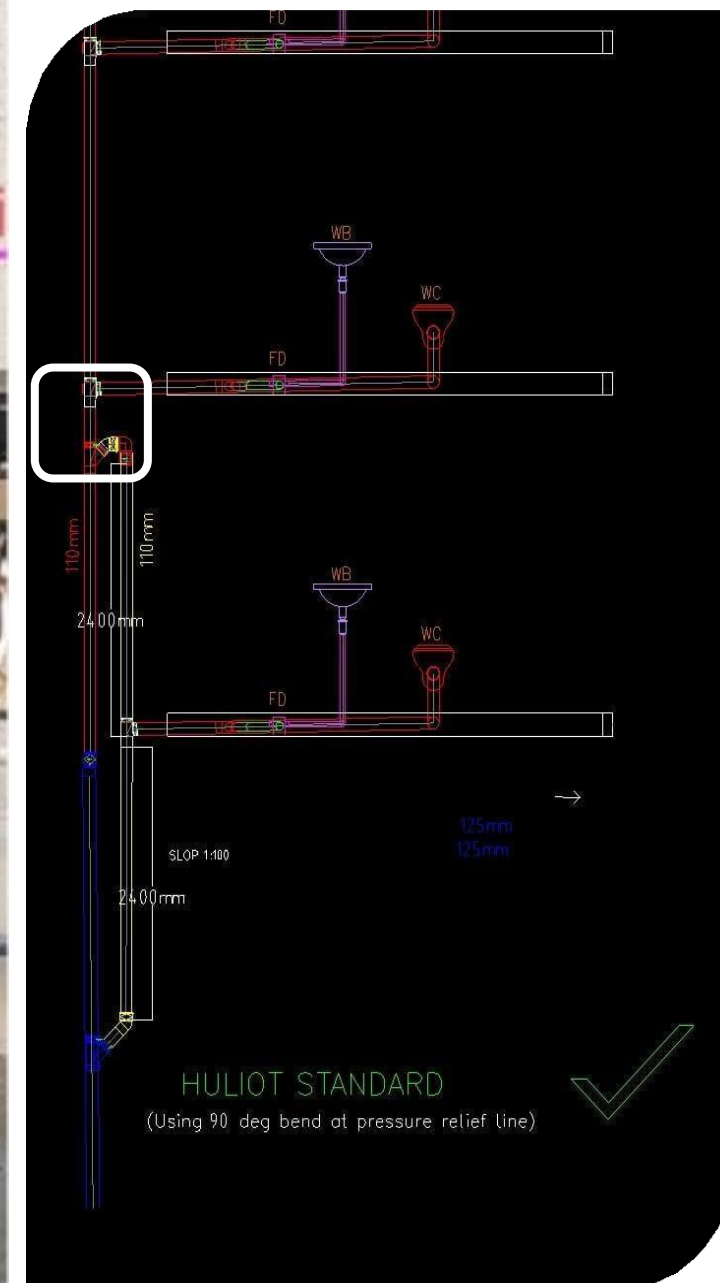
Reason:

A minimum straight pipe section allows smooth wastewater flow and proper air movement before the change in direction. Diverting the line immediately after a **fitting** can disturb the flow and create turbulence.

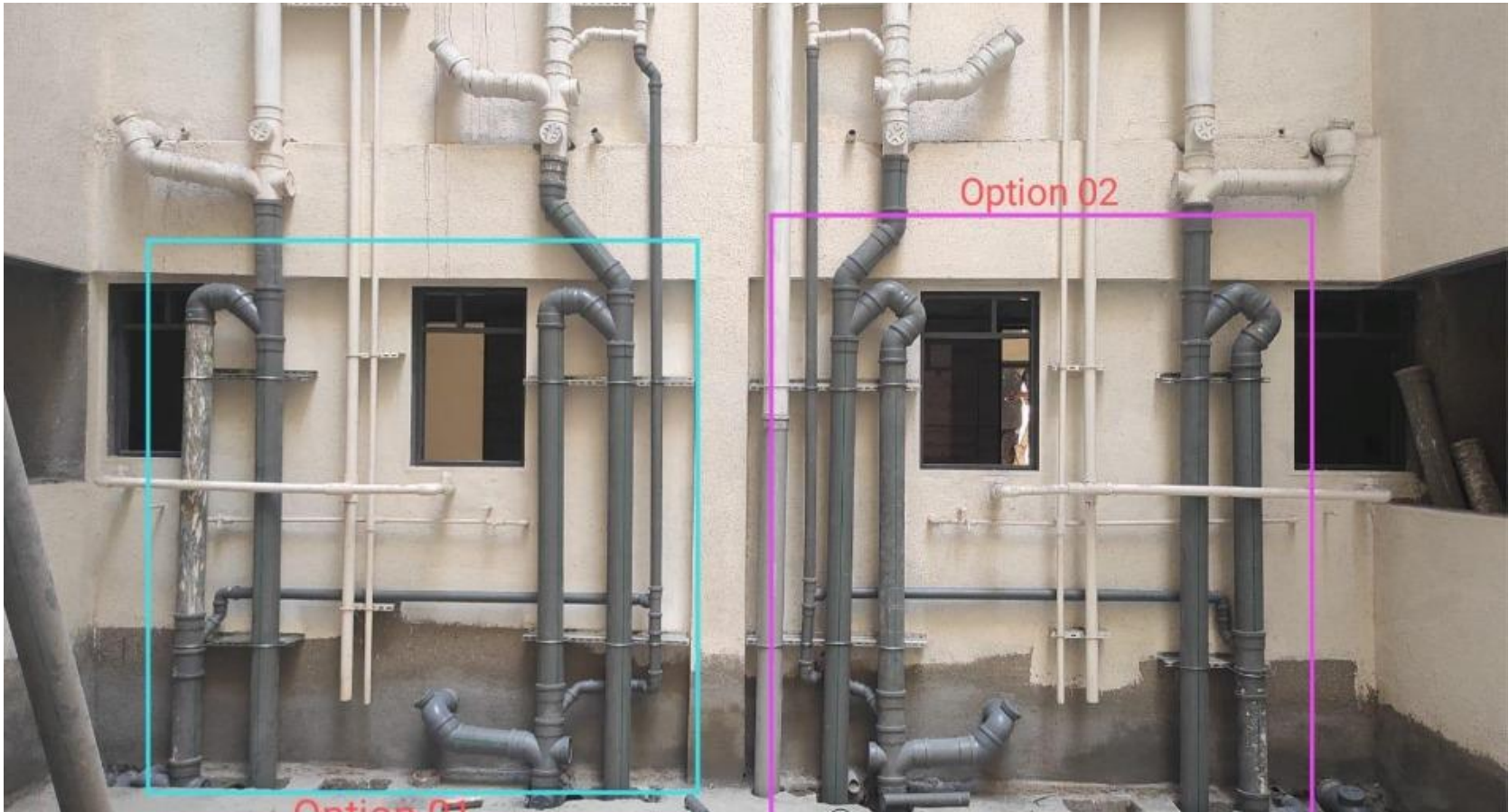
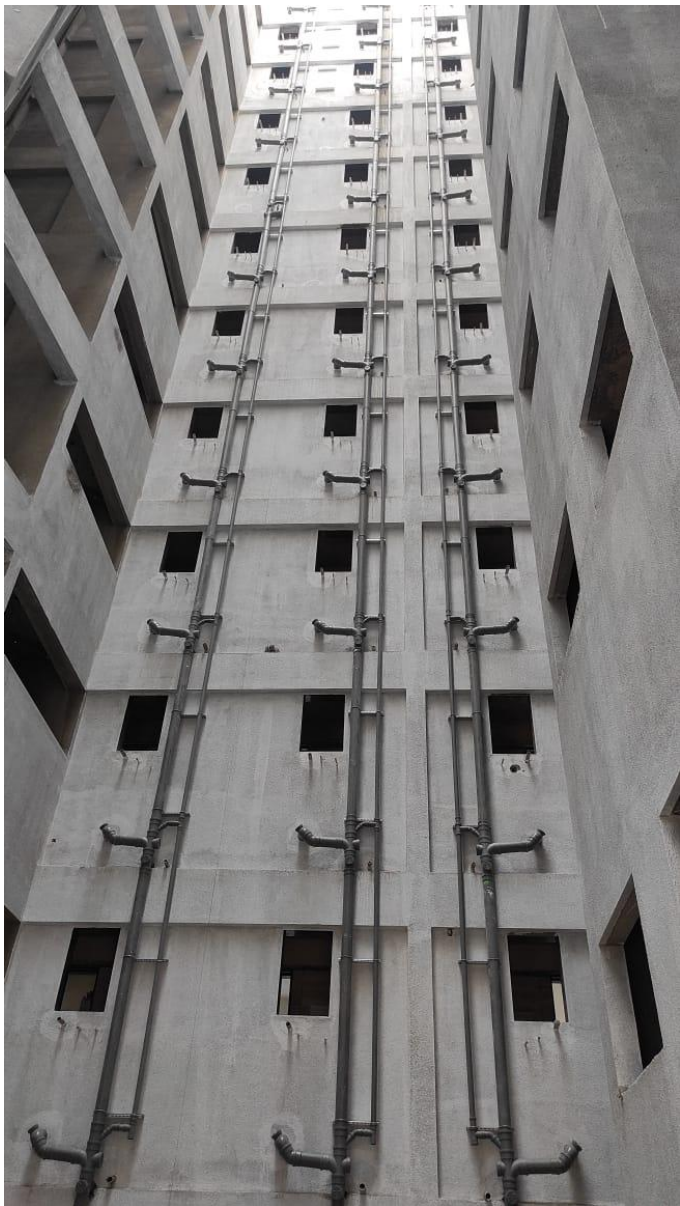
Possible Issues if Not Maintained:

- Backflow to the upper floor fixtures.
- Increased risk of trap seal disturbance.
- Reduced overall drainage system performance.

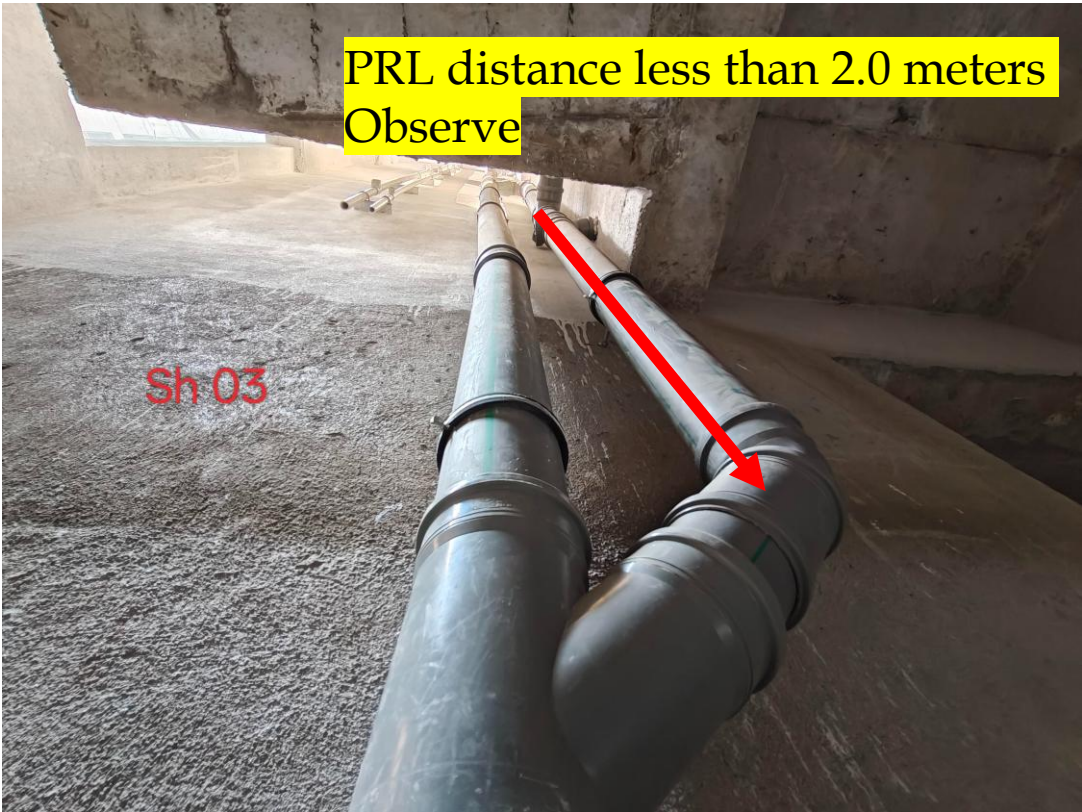
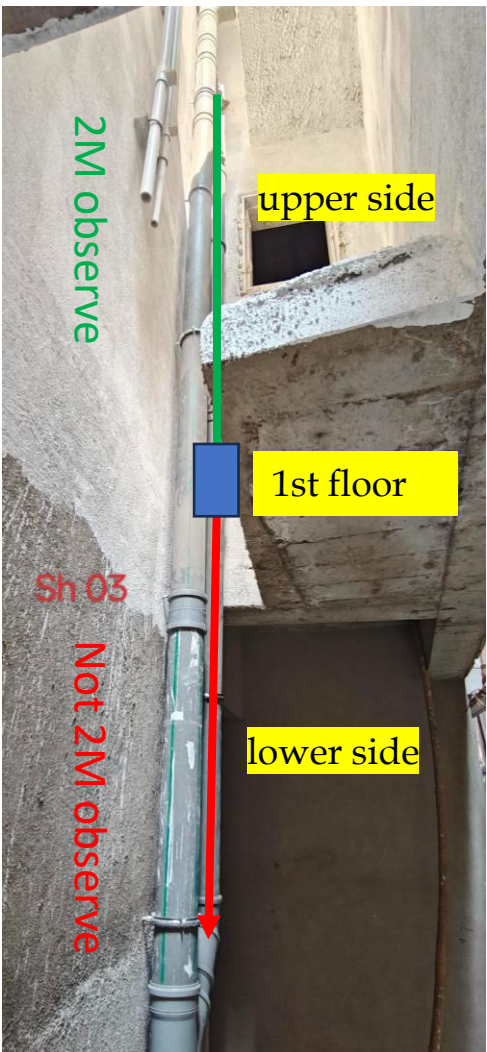
Maintain a minimum 250-300 mm straight pipe length before any diversion (bend/offset). Failure to maintain this distance may result in backflow issues on the respective floor and improper drainage performance.



One Pipe System PRL installation Reference Photo



PRL Mockup Observation: PRL has two parts at the 1st floor level, one connection above and one connection below. As per requirement, minimum pipe length required is 2.4 m (minimum acceptable 2.0 m). At the upper side of the 1st floor PRL connection, 2.0 m length is observed, which is acceptable. However, at the lower side of the 1st floor PRL connection, the length observed is less than 2.0 m, which is not as per requirement. Correct it immediacy as explained.



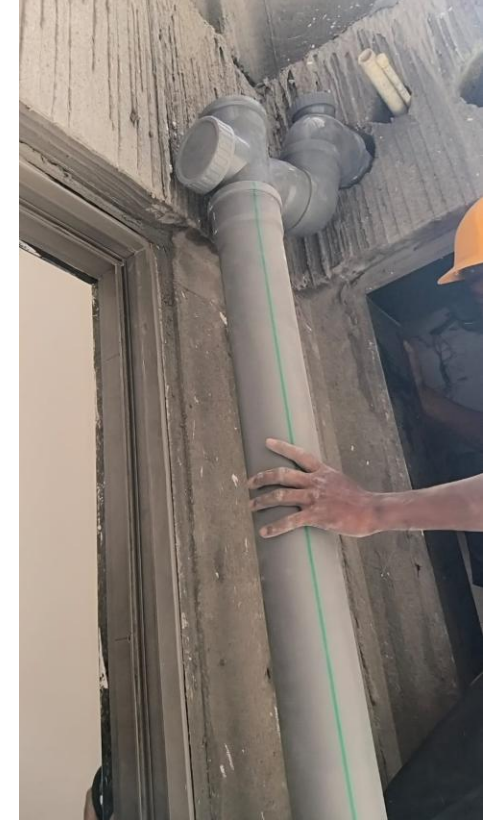
The observed trap-to-trap connection is not in accordance with standard plumbing practice, as each trap should discharge independently into a properly vented branch drain or vertical stack. The current arrangement, where the drainage line is reduced from 75 mm to 50 mm and another trap is connected downstream, is not permitted. Trap-to-trap connections and reduction of pipe diameter in the direction of flow can lead to siphonage, blockage, backflow, and failure of the trap water seal. Connecting multiple traps to an undersized line increases the risk of choking and water seal failure, making this arrangement unsuitable and requiring rectification.



During our recent site visit, we observed that in the bathroom drainage rough-in, a fixture branch trap has been connected to the outlet of the floor trap, creating a double trap (trap-to-trap) condition. This installation is not in compliance with IS 1742:1983 Clause 5.3 and NBC 2016 Part 9 Clause 3.5.3, which require each fixture trap to discharge independently and directly into the branch drain — not into another trap. A double trap creates a sealed air pocket between the two water seals, which will inevitably lead to trap siphonage, foul odour complaints, and drainage performance issues after occupation. We kindly request that the contractor rectify this before waterproofing proceeds, and that a post-correction photograph be shared with us for records. We are available to guide the correct installation detail if required.

In a single stack system, the use of swept fittings is recommended. Existing Y-fittings should be replaced with swept fittings to ensure proper flow and compliance with system standards.

- **In a single stack drainage system, only SWIFT fittings are recommended as per standard practice. The use of Y-fittings is not advised, as they can adversely affect airflow and pressure balance within the stack.**



Drainage Installation & Testing Checklist

(For Proper Installation, Quality Control & Handover)

A. Drawing & Approval Compliance

1. Drainage pipe installation carried out strictly as per approved drawing and site-in-charge instructions

☐ Yes ☐ No

2. Pipe routing followed as per approved drawing or approved site modification

☐ Yes ☐ No

3. Any deviation from drawing approved by consultant / site-in-charge

☐ Yes ☐ No ☐ Not Applicable

B. Pipe Installation & Support

4. Pipe diameter, type, and fittings installed as per specification

☐ Yes ☐ No

5. Pipe supports / clamps provided as per manufacturer recommendation and standard spacing

☐ Yes ☐ No

6. Rigid support provided for pipes above 600 mm length (MS / GI support where required)

☐ Yes ☐ No ☐ Not Applicable

7. Trap and trap arm properly supported and firmly fixed

☐ Yes ☐ No

8. Proper slope maintained throughout horizontal drainage lines

☐ Yes ☐ No

C. Jointing & Termination

9.All joints properly connected and sealed as per installation guidelines

☐ Yes ☐ No

10.Open pipe ends properly closed with end caps during installation and testing

☐ Yes ☐ No

D. PRL / Vertical Line Compliance (If Applicable)

11.PRL created separately at last floor level

☐ Yes ☐ No ☐ Not Applicable

12.After 2nd floor connection, no 45° shoe connected directly

☐ Yes ☐ No ☐ Not Applicable

13.Minimum 300–500 mm straight pipe provided after 2nd floor connection before PRL start

☐ Yes ☐ No ☐ Not Applicable

E. Water Testing

14.Drainage water testing carried out for each toilet

☐ Yes ☐ No

15.Water testing height maintained at minimum 500 mm or 1.0 meter as per site requirement

☐ Yes ☐ No

16.Proper end caps used during water testing

☐ Yes ☐ No

17.Any leakage observed during water testing

☐ Yes ☐ No

18.Leakage rectification carried out immediately and re-testing done

☐ Yes ☐ No ☐ Not Applicable

F. Final Readiness Before Finishing

19. Inside traps cleaned properly and free from cement mortar before fixing grating / jali / shower channel

☐ Yes ☐ No ☐ Not Applicable

20. Drainage line cleared and ready for waterproofing / finishing works

☐ Yes ☐ No

G. Documentation & Sign-off

21. Checklist maintained separately for each toilet / shaft

☐ Yes ☐ No

22. Checklist verified and signed by site supervisor / engineer

☐ Yes ☐ No

Work done as per drawing . **Not follow standard installation practice stop the work immediately** and do the rework as per suggested or drawing.



Please carry out the vertical mock-up to check future clearance.
Ensure adequate space is maintained for all services to avoid any obstruction at a later stage.

Core opening at site is not as per approved drawing. Due to this mismatch, two additional 45° shoes have been used to maintain proper alignment of the pipe.
Technically, this does not create any functional issue and flow will not be affected.
All other work has been executed as per approved drawing.



General Point

- Open pipe ends and trap top needs to close to avoid any cement slurry , cement mortar not goes inside trap. Trap needs to clean and water seal needs to fix properly before fixing zali to avoid any foul smell issue later stage.
- All pipes and fittings must be fixed at the correct angle.
- If the outlet pipe is not taken out straight and perfectly vertical, the vertical fittings cannot be properly pushed or seated. This can lead to improper jointing and may cause leakage issues. Ensure the outlet pipe is aligned straight and vertical from the starting point before installing any fittings.
- Do the vertical mockup Fatcher clarence.
- Sunken pipes must be properly fixed with cement mortar to ensure the alignment does not change during waterproofing works.

PVC and PP are different materials and follow different jointing methodologies. Mixing both systems is not recommended as it may lead to improper joint performance and long-term leakage issues. Replace the PVC fitting with a compatible Huliott PP fitting to ensure system integrity and proper installation.

PVC and PP are different materials and have different joint systems. So mixing both is not recommended. Please replace it with a Huliott fitting.



Duvalle grand Good Clamping





Swept Tee fittings are used in vertical single stack systems to maintain annular flow, preserve the central air core, control pressure fluctuations, and protect trap water seals. Y fittings are unsuitable for vertical stacks due to turbulence and airflow disturbance

Use Swept Tee fittings instead of Y fittings in vertical single stack systems to maintain proper airflow and protect trap seals.”



PRL (Pressure Relief Line) stabilizes air pressure in a single stack drainage system by relieving pressure fluctuations caused by high-velocity wastewater, thereby protecting trap water seals and ensuring system reliability in high-rise buildings.

PRL (Pressure Relief Line) controls positive and negative air pressure generated by high-velocity discharge in single stack systems, preventing trap seal loss in high-rise buildings.

PRL (Pressure Relief Line) is a critical component of the single stack drainage system that stabilizes positive and negative air pressure created by high-velocity wastewater discharge, relieves pressure fluctuations due to terminal velocity in high-rise buildings, and protects trap water seals, thereby ensuring safe, odor-free, and reliable system performance.

Work was not executed as per drawing and was instead carried out according to site conditions. At site, a 90° bend was used below WC and washbasin . We recommend using two 45° shoes instead, as a 2×45° configuration provides smoother flow and reduces noise, which is especially important in a suspended drainage system.

Huliot always checks that installations are done as per approved drawings. Any deviation due to site conditions must be approved by the concerned site authority before implementation.



Fix the pipes properly with **cement mortar** so that the **pipe alignment** does not change during waterproofing work.



Work done as per drawing.

Work not done as per drawing.

Work done as per site condition changes.

Installation has been completed as per the drawings.

The contractor changed the washbasin pipe routing with approval from the site team. The change is technically acceptable

The mock-up has not been executed properly. At the site, a 90° bend was used below the WC and washbasin. We recommend using two 45° shoes instead, as a 2 × 45° configuration provides smoother flow and reduces noise, which is especially important in a suspended drainage system. Pipe alignment is not proper, clamps have not been provided, trap support is missing, and fittings are installed inside the core.

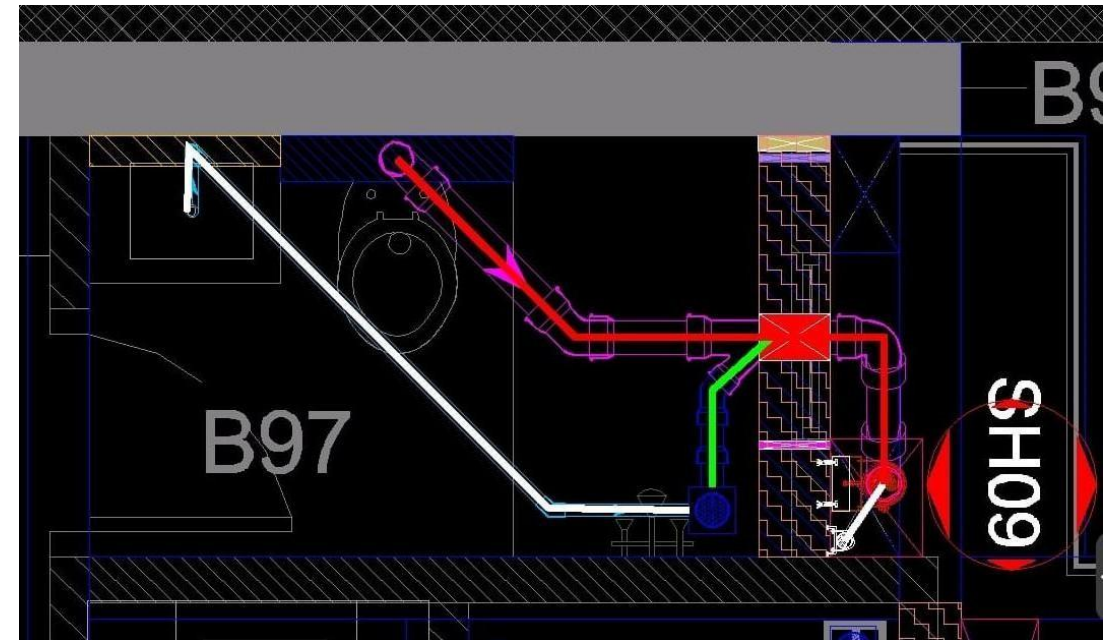
Pipe routine done as per drawing.

Huliot checks all installations to ensure they are executed as per approved drawings. Huliot always recommends that all installations be carried out strictly according to these drawings. If any site condition requires a change, including pipe routing modification, prior approval must be obtained from the concerned consultant before execution.

Huliot verifies installations as per approved drawings. Any deviation, including pipe routing changes, requires prior consultant approval before execution.

Reference photo





Installation has been completed as per the drawings.

In the Shaft outlet pipe is not straight ,vertical fittings will not seat properly, leading to poor joints . fix outlet pipe straight . before waterproofing.



vertical mockup – checking clearance of connection obstacles,
you need to verify that the pipe connection has proper space
from slab, beam, wall, and other services.





This type of situation occurs when the vertical line is not aligned straight on each floor. Due to misalignment at every level, the pipe does not continue in a straight line.

Providing a 45° shoe at the required location will help maintain proper alignment and ensure the vertical line runs straight. But it not stander practise

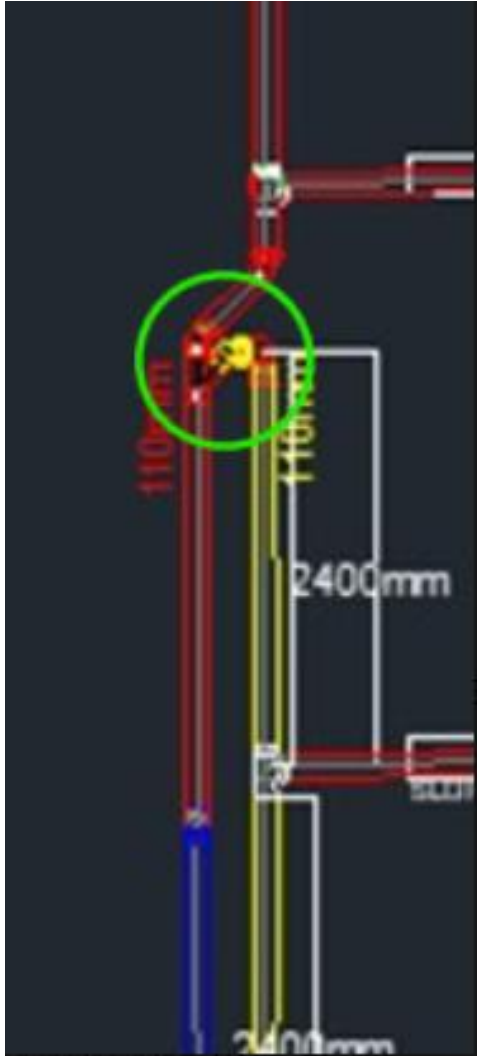


Use two plain sweep tees in the vertical line instead of a door fitting. It is recommended to provide an inspection pipe at alternate floors for vertical inspection and maintenance access



PRL connection must start with a Y-Branch fitting. Swept Tee is not recommended at this point

At the starting point of the PRL, a Y-Branch fitting should be provided. Swept Tee should not be used at this location.



In PRL Use Y branch Instead of Sweep t.

Kindly ensure a clear gap of 15–20 mm is maintained between the slab and the fitting. Request not to insert the fitting inside the core, as this will help facilitate smooth future maintenance.



The 110mm drainage pipe is installed with excessive slope.

The slope shall be corrected and maintained at **1:100** as per standard requirement to ensure proper flow and avoid operational issues





Blockage

Yes, we can guide you.

First, please confirm where the blockage is located. Is it in the wash basin line, WC line, floor trap, vertical stack, main header line, or chamber?

The method to remove concrete blockage depends completely on the location and severity. Treatment is different for each case.

Please share:

- Exact location
- Pipe diameter
- Whether the pipe is vertical or horizontal
- Approximate length of blockage

Based on that, we can suggest the correct procedure.

- Most of the time, if it is a small blockage in a wash basin line, you can use a drain cleaner. You may use any standard brand available locally. Make sure to flush properly with water after treatment.
- For WC blockage, use a plunger first. In many cases, this clears minor obstruction effectively.
- For vertical pipes and chamber blockages, use rodding with a proper flexible rod. If the concrete is hard and fully set inside the pipe, mechanical cleaning may be required.



Stainless Steel Drain Cleaner Wire Stick



Big Chamber



Drain cleaner



Flush Toilet Plunger

Cement has entered and blocked the trap. Since the cement has hardened, it cannot be cleared using chemicals. The trap needs to be cut and replaced to ensure proper flow and maintain the water seal. Please arrange for rectification at the earliest.

Trap is blocked due to hardened cement. Cleaning is not possible. Trap replacement is required. Kindly take immediate action.

- If Cement Is Still Fresh (Soft Stage):

Flush immediately with water and clear using flexible rod/wire before it sets.

- If Cement Is Hardened Inside the Trap:

Chemical cleaners will not work. Mechanical action or replacement is required.

- Remove and Replace (Best Option):

Cut the blocked trap and install a new one to ensure proper flow and water seal.

- Mechanical Breaking (Only If Minor Block):

Carefully break the cement using tools. Risk of trap damage is high.

- If Cement Blocked Vertical Pipe Also:

Carry out proper rodding. If fully hardened, cut and replace the affected pipe section.

- ✓ Checking if material is approved
- ✓ Supporting technical decisions in front of consultant
- ✓ Giving strong justification in reports
- ✓ Avoiding site disputes (standard-based discussion)

Perfect report format (use this)

- ◆ **Observation:**

What is wrong

- ◆ **Standard Reference:**

Mention IAPMO / code

- ◆ **Technical Impact:**

Why it is a problem

- ◆ **Recommendation:**

What to do

Final Simple System (Remember This)

Whenever you inspect anything, follow:

 **CHECK** →  **REFER** →  **REPORT** →  **CONTROL**

Short formula for you

 **No standard = No approval**

Huliot → EN Standard (What it means)

Huliot Pipes drainage systems are designed as per European norms like:

Main Standards

- **EN 1451**

-  PP (Polypropylene) drainage systems inside buildings

- **EN 12056**

-  Design rules (slope, venting, stack system)

Simple understanding.

- **EN 1451 = Material standard (pipe & fittings quality)**

- **EN 12056 = Design standard (how to install system)**

Example (your real case)

Single stack vs two stack:

Say:

“Huliot proposes single stack system based on EN 12056, which is hydraulically designed.

Random modification may affect performance.”

3. Strong Report Justification



Use this format:

Observation:

Installation not as per Huliott recommended system.

Standard Reference:

System is designed as per EN 12056 & EN 1451.

Impact:

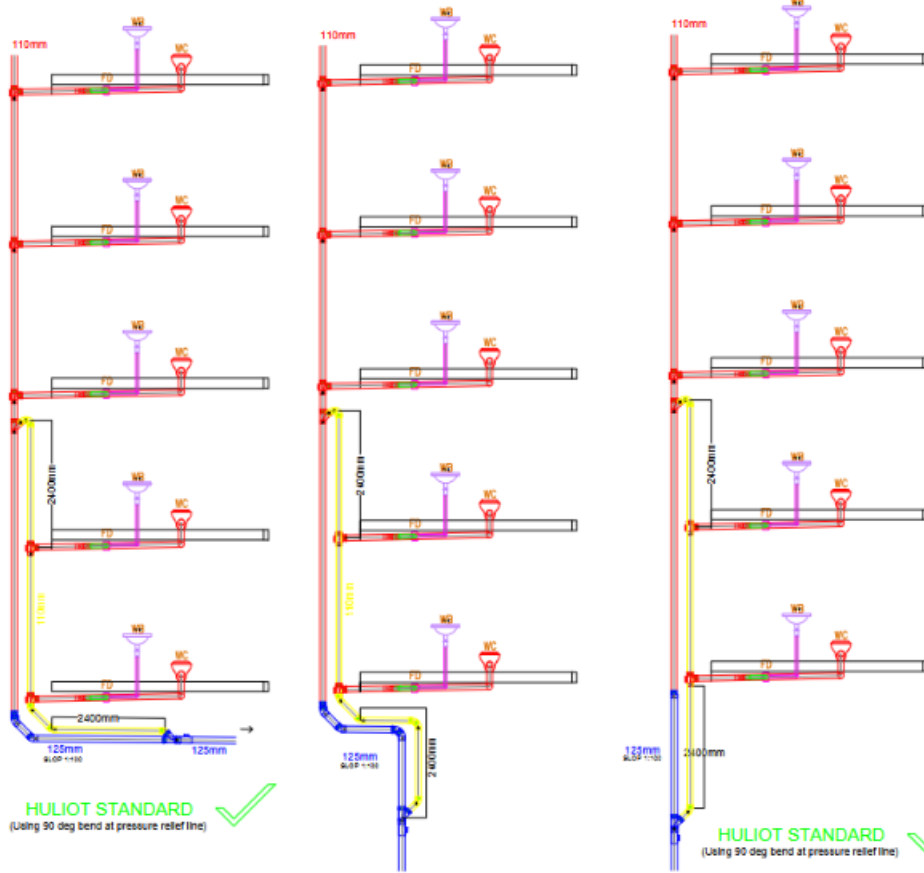
Improper flow / noise / trap seal loss.

Recommendation:

Follow Huliott design & approved drawings only.

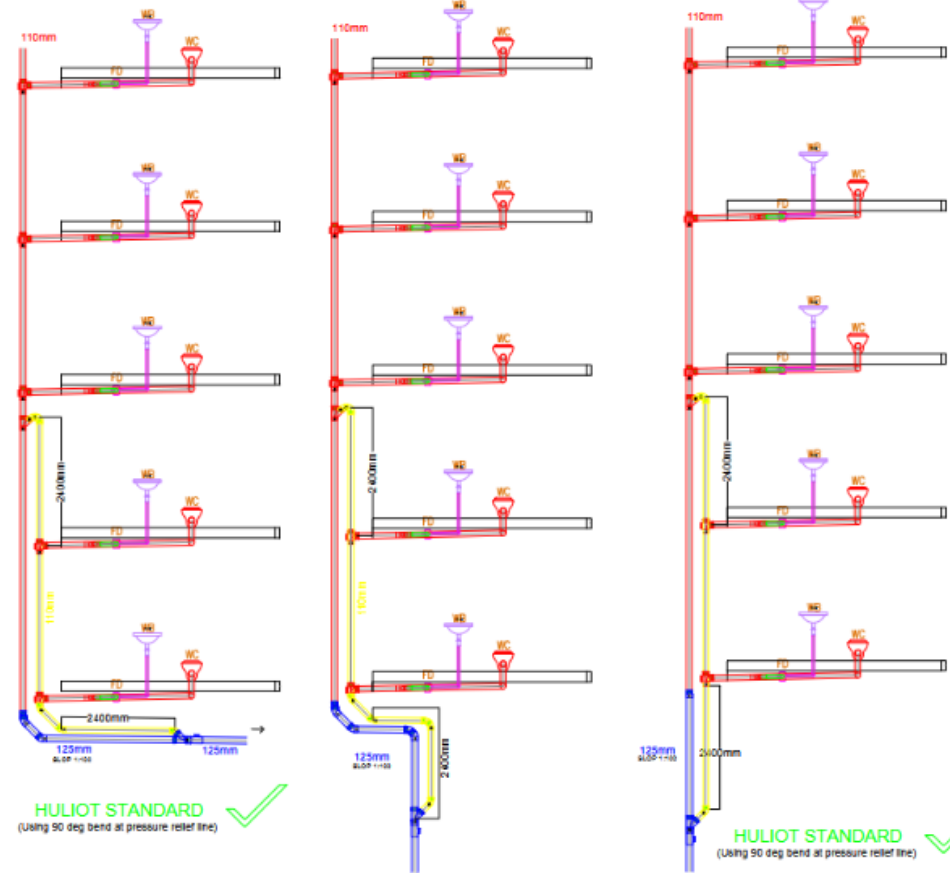
“Installation deviates from EN standard-based system design, hence not acceptable.”

Suspended with Huliot Swept Tee system



HULIOT STANDARD
(Using 90 deg bend at pressure relief line)

Suspended with Huliot Swept Tee system

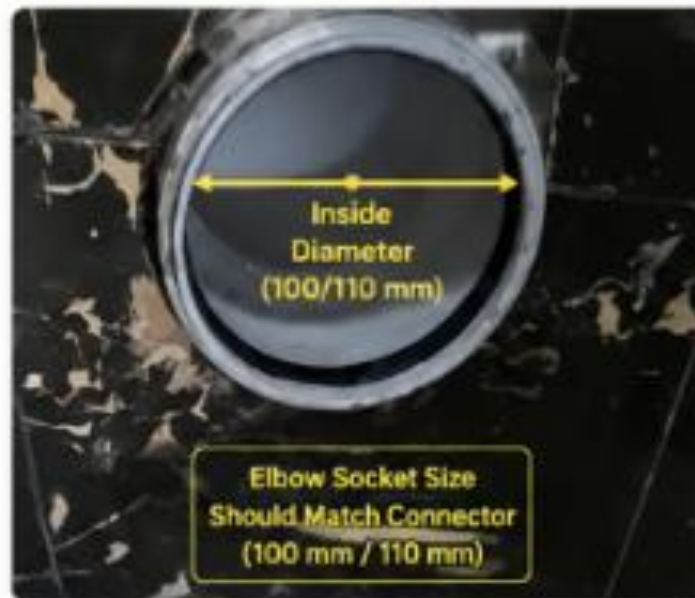


HULIOT STANDARD
(Using 90 deg bend at pressure relief line)

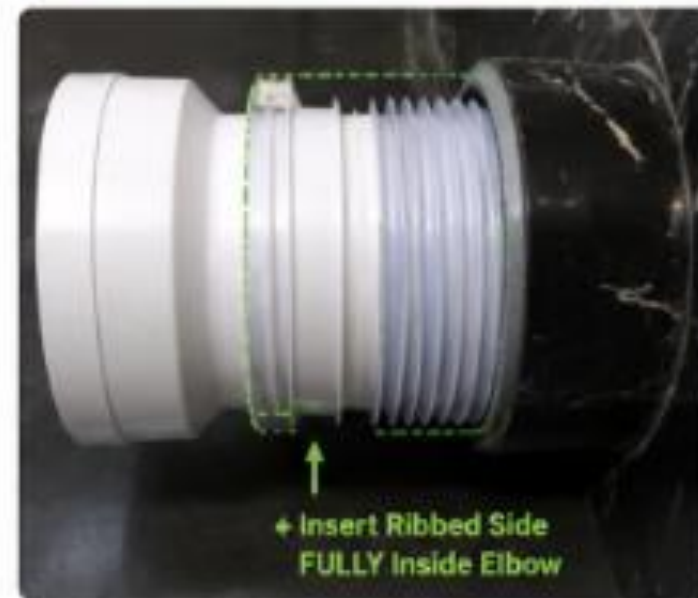
1. PAN CONNECTOR

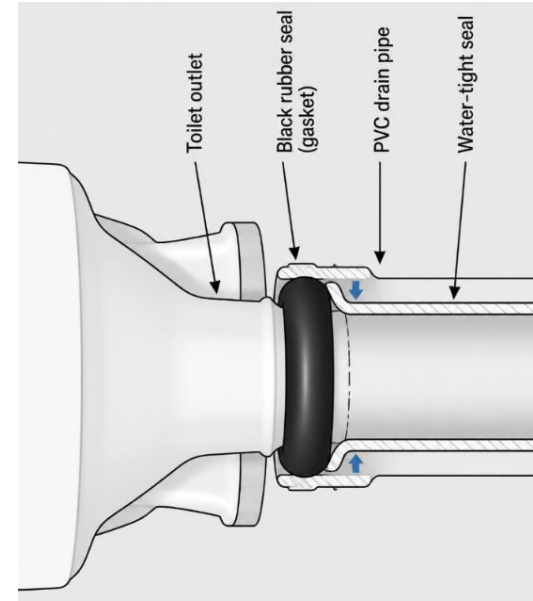


2. WC ELBOW (INSIDE)



3. CONNECTION (SIDE VIEW)





Purpose of the Black Rubber Seal (Main Point):

- It works like a “washer” or “O-ring”.
- Creates a **water-tight seal** between the toilet outlet and the PVC pipe.
- Prevents **leakage** from the joint.
- Allows slight movement so the pipe doesn't crack if there is any settlement.

<https://www.youtube.com/watch?v=qaeradNMqiQ>

Huliot MFT Comparative			
	MFT with Socket	MFT 7 inch	MFT Slovenia
Trap			
Top Dia	110	110	110
Inlet Dia	50	50	50
Inlet Nos	3	3	3
Outlet Dia	75	75	75
Height	245	173.5	173
Trap Seal	50 mm	50 mm	50 mm
Flow Rate	60 LPM	60 LPM	60 LPM
Socket Provided	Yes	No	Yes
Height Riser Required	No Plain Pipe to be pushed inside the socket	Yes	No Plain Pipe to be pushed inside the socket
Application	Underslung System	Sunken System	Underslung & Sunken Both

Use height riser above the multi-trap instead of pipe piece



height riser shall be used above the multi-trap instead of a cut pipe piece.



Clamp need to fix close to the pipe socket as marked area.













